

Teacher Guide — middle / module-05-dot-addition

IKS Curriculum

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Overview

Module 5 — Bindu Paddhati (Dot Method of Addition)

Age band: Middle · **Grades:** 6–8 · **Duration:** 10 sessions × 45 min

Learning objectives

By the end of this module, students will be able to:

1. **Recall** the Bindu Paddhati (“dot method”) procedure for column addition: scanning each column for groups that sum to 10, marking dots, and counting dots as the carry.
2. **Execute** column addition on 3- to 4-digit numbers with at least three addends using the dot method, in under half the time their standard-algorithm baseline would take.
3. **Explain** why the dot method works — that a dot represents a multiple of 10 carried to the next column.
4. **Apply** the method to mental addition of 2-digit numbers using the “completing the ten” pattern (e.g. $19 + 8 = 19 + 1 + 7 = 27$).
5. **Compare** the dot method with the standard right-to-left carry algorithm and articulate when each is faster.

NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems (Vedic mathematics) with mainstream arithmetic.
- **§4.23** — Experiential learning via paired drills and timed challenges.
- **§4.27** — Local context: drawing on regional Vedic-math pedagogical traditions.
- **§5.1** — Foundational numeracy and fluency.

Prerequisites

- Confident with addition of two single-digit numbers up to 18.
- Comfortable with carrying in standard column addition.
- Can write numbers in columns aligned by place value.

Key vocabulary

See `_shared/glossary.md`.

- *bindu* — dot
- *paddhati* — method
- *Sankalana Vyavakalanābhyām* — “by addition and by subtraction” (7th of Tirthaji’s 16 sūtras); related but not the dot method per se
- *Pūraṇāpūraṇābhyām* — “by completing or non-completing” (8th sūtra); names the “complete the ten” idea

Materials checklist

- A4 sheets with pre-printed columns of numbers (one per student, varied difficulty)
- Stopwatches or phone timers for paired drills (1 per pair is enough)
- A class wall-poster of “pairs that make 10” (1+9, 2+8, 3+7, 4+6, 5+5)
- Whiteboards or scrap paper for live demos

Day-by-day overview

Day	Title	Focus	Assessment
1	The Hook — How fast can you add?	Baseline timed test; reveal dot method	Self-timed baseline
2	Making Tens	Pattern recognition: pairs to 10	Pattern quiz
3	Single Column with Dots	Procedure with 1-digit addends	Formative quiz
4	Multiple Columns	2-, 3-, 4-digit numbers with carries	Activity rubric
5	Mid-module Check	Speed test + formative	Formative quiz
6	Why It Works	Mathematical reasoning behind the method	Reflection
7	Mental Math Patterns	“Numbers near 10” technique	Drill challenge
8	Project Session 1	Plan + start	Project draft
9	Project Session 2	Build + rehearse	Project draft
10	Final Assessment	Speed test + presentation	Summative + project

Citation sources used

- **SutraGanita corpus** (/Users/sree/Projects/SutraGanita/content/) — primary references in `sources.md`:
 - *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — §5.2 Mental Addition, §5.3 Completing the Whole, §5.4 Columns of Figures
 - *Microsoft Word - sutras1.pdf* — list of 16 Tirthaji sūtras
 - *Vedic_Maths.pdf* (ISKCON Desire Tree) — practice problem sets

Project brief (one-line teaser)

Each pair makes a 60-second teaching video, a printed workbook, OR a comparative speed-test data project showing the dot method to a Grade 4 audience. Full brief in `assessments/project-brief.md`.

10-Day Lesson Plan

Module 5 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Hook — How fast can you add?	<code>days/day-01-hook.md</code>	Baseline timed test
2	Making Tens	<code>days/day-02-making-tens.md</code>	Pair drills with the “10-pair” cards
3	Single Column with Dots	<code>days/day-03-single-column.md</code>	Formative quiz
4	Multiple Columns	<code>days/day-04-multiple-columns.md</code>	<code>activities/activity-01-column-drill.md</code>
5	Mid-module Check	<code>days/day-05-mid-module-check.md</code>	<code>quizzes/formative.md</code> Part B
6	Why It Works	<code>days/day-06-why-it-works.md</code>	Reflection paragraph
7	Mental Math Patterns	<code>days/day-07-mental-patterns.md</code>	<code>activities/activity-02-near-base.md</code>
8	Project Session 1	<code>days/day-08-project-session-1.md</code>	Project draft
9	Project Session 2	<code>days/day-09-</code>	Project draft

Day	Session title	Lesson file	Activity / assessment
10	Final Assessment	project-session-2.md days/day-10-final-assessment.md	quizzes/summative.md + project

Pacing notes

- Day 1 establishes a baseline speed (each student times their own standard-algorithm addition of a 4×4-digit sum). Day 5 retests the SAME problem set with the dot method; the visible speedup is the module’s motivation.
- Days 2–4 are the core. Compressed schedule: combine Days 2+3 into one period if needed.
- Day 6 (“Why it works”) can be pushed to Day 7+ if the class is enjoying the drills.
- Project (Days 8–10) is shorter than Panchabhuta’s because the artefact options are more constrained.

Daily warm-up (recurring)

Every class starts with **1 minute of “make-10 fluency”**: teacher calls out single digits; students call back the partner that makes 10. (8 → 2, 6 → 4, 9 → 1, etc.) This is the foundational skill.

What students leave the module with

- Fluency at scanning a column of numbers and finding 10-pairs.
- Confidence adding 4-digit, 5-addend columns in their head or with paper.
- Awareness that “Indian Math” is not a single ancient block — it includes 20th-century pedagogical innovations like this one.
- One practice artefact (their project) they can share with younger students.

Teacher Notes

Teacher Notes — Module 5 Dot Addition (Middle)

Posture

This module is about **speed and pattern recognition**, not about memorising a Sanskrit name. The win is when a student instinctively scans a column of numbers and groups them into 10s without conscious thought. The Sanskrit term *Bindu Paddhati* is the cultural anchor; the *technique* is what they take home.

Honesty about provenance: the dot-column method described in this module is a 20th-century pedagogical technique. It draws on the same instinct as Tirthaji’s sūtras *Sankalana Vyavakalanābhyām* and *Pūraṇāpūraṇābhyām*, but it is not itself one of the 16 sūtras. Don’t

oversell it as “ancient Vedic math.” At Senior band students unpack this distinction in detail; at Middle band just don’t claim what isn’t true.

Common misconceptions

1. **“This method is slower because I have to write more (the dots).”** True for some students initially. Practice fixes it — by Day 5 the dot method should be at least 50% faster on multi-addend columns.
2. **“I only need to look for exact 10s — 9+1, 8+2, etc.”** Also accept $9+2 (= 11)$ — put down 1, carry one dot. The point is to *group efficiently*, not to find exact 10s only.
3. **“The dot is the same as the carry digit.”** Yes — each dot represents one “1” carried. If you have 3 dots in a column, you carry 3. Reinforce this on Day 6.
4. **“This is just regrouping in disguise.”** Partly true — and that’s fine. It’s a *visual* technique for regrouping that exploits the fact that scanning beats serial addition.

Differentiation hints

- **Tier up:** Skip dots entirely and do the running total in your head. Faster students should be doing fully mental column addition by Day 7.
- **Tier down:** Use coloured pens — different colour for each 10-group identified in a column. The colour scaffolds the “scan and group” action.

Differential pacing

- A class with strong arithmetic baseline (most students adding 4-digit + 4-digit in <30 sec) can compress the module to 6–7 days by combining Days 1+2 and Days 4+5.
- A class with shaky baseline addition should NOT skip Day 2. The “make-10” fluency is the prerequisite for everything else.

Sensitivity notes

- Some students are anxious about timed math. Be clear that the timed activities are **personal baselines** — students race themselves, not each other. Public leaderboards are *not* used.
- Avoid the framing “the West has standard algorithm, India has the fast method” — both methods are taught in Indian schools, both are *useful*, and the standard algorithm is itself partly of Indian origin via Brahmagupta. Module 6 (future, “Astronomy”) will touch on this.

Project oversight

- Three project formats (60-sec teaching video, printed workbook, comparative-speed data project) each have different production challenges.
- Video projects: check before Day 8 that the pair has access to a phone with a camera AND a quiet place to record. Otherwise redirect.
- Speed-test data projects: students collect data from 5–10 family members or schoolmates. They time both methods. Confirm by Day 9 that they actually have the data — otherwise the project collapses on Day 10.

Citation discipline

Citations for this module are PDF references (e.g. *Vedic Mathematics Batch1.pdf*, §5.4), not scripture references. Students should cite the PDF name and section in their projects. This is the standard for math modules in this curriculum.

If you only have 6 days

- Day 1 (hook + baseline)
- Day 2 (combined: making tens + single column with dots)
- Day 3 (multiple columns)
- Day 4 (mid-module check + mental patterns combined)
- Day 5 (one project session)
- Day 6 (final assessment, no project presentation)

Sources

Sources — Module 5 Dot Addition (Middle)

The **vidya-karana RAG** does not cover Vedic-math sources, so this module draws directly from SutraGanita's source corpus (`/Users/sree/Projects/SutraGanita/content/`). All PDFs cited below are accessible in that directory.

Primary source (used most heavily)

Vedic Mathematics Batch1.pdf — *Vedic Cultural Center (ISKCON), 1420 228th Ave SE, Sammamish WA 98075*

- **§5.2 — Mental Addition** (page 16). Pattern recognition: “if you need $6+7+4$, see that 6 and 4 make 10, then add 7 to get 17.”
- **§5.3 — Completing the Whole** (page 17). Extends to bigger numbers: “for $19+8+1$, see $19+1$ makes 20, then add 8.”
- **§5.4 — Columns of Figures** (page 18). The dot/scanning method: scan a column for pairs summing to 10, put the units digit down, carry 1 (the dot count).
- **§5.5 — By Addition and By Subtraction** (page 19). Numbers near a base (9, 19, 99): add the base, take away the difference. E.g. $33+9 = 33+10-1 = 42$.

Cite as: *Vedic Mathematics Batch1.pdf*, §5.X.

Supporting sources

- **Microsoft Word - sutras1.pdf** — single page listing all 16 Tirthaji sūtras and their 13 sub-sūtras. Used in Day 6 (“Why It Works”) to give the historical context of the named sūtras *Sankalana Vyavakalanābhyām* (#7) and *Pūraṇāpūraṇābhyām* (#8).
- **Vedic_Maths.pdf** (ISKCON Desire Tree) — workbook style; contains §“Addition – Dot Method” (Session III) practice sets. Used for the drills in Day 4 and Day 7.

- **FUNDAMENTAL AND VEDIC MATHEMATICS .pdf** — comprehensive 65-page treatment; useful as teacher reference; not directly cited in student-facing material.

Historical context

- **Tirthaji, Bharati Krishna.** *Vedic Mathematics* (1965), Motilal Banarsidass. — The 16-sūtra system. Note: the dot-column method described in §5.4 of the source above is *not* one of Tirthaji's sūtras; it is a pedagogical technique often taught alongside the sūtras. Be honest about this distinction with Senior students; at Middle band a one-line note is enough.
- **Plofker, Kim.** *Mathematics in India* (2009), Princeton University Press. — Historical context of pre-Tirthaji Indian mathematics; mentions Bakhshali manuscript addition methods (~7th century CE).

What to tell students about the citation chain

The “dot method” is not directly from any single classical Sanskrit source — it is a teaching technique developed in 20th-century Indian mathematics pedagogy, drawing on the same pattern-recognition spirit as Tirthaji's sūtras *Sankalana Vyavakalanābhyām* and *Pūraṇāpūraṇābhyām*. At Middle band, we'll cite the modern source (Vedic Cultural Center course material). At Senior band, the historical question itself becomes part of the lesson.

Day 1 — The Hook: How Fast Can You Add?

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will record a personal baseline time for adding a 4-addend × 4-digit column using the standard algorithm, and will have seen the dot method solve the same problem visibly faster.

Materials

- Printed baseline sheet: ONE column-addition problem (e.g. $5728 + 3417 + 6892 + 4536 = ?$). One per student.
- Stopwatch or wall clock with second hand. (One per pair is plenty.)
- Whiteboard / large paper to demonstrate.

Vocabulary introduced today

- *Bindu Paddhati* (IAST: bindu-paddhati; lit. “dot method”) — the technique we'll learn.

Lesson flow

Warm-up (5 min)

- Teacher: “How long does it take you to add four 4-digit numbers? Make a guess.”
- Take 3–4 guesses. Write the range on the board (e.g. “30 sec to 2 min”).

Baseline timing (10 min)

- Hand out the baseline sheet face-down.
- Students write their name and predicted time on the back.
- On “Go,” flip and solve using whatever method they normally use (standard algorithm).
- Record their actual time (the wall clock is running).
- Mark answer; **honest self-correction** — they’re racing themselves over the module, not each other.

Teacher demonstration (10 min)

- Write the same problem on the board.
- Solve it using the **dot method**, narrating:

5728
3417
6892
4536+

Units column (top to bottom): 8, 7, 2, 6.

- Scan: $8 + 2 = 10$. Mark a dot above the 2. Running total: 0.
- Continue: 7. Running total: 7.
- Continue: 6. Running total: 13. That’s a 10 — mark a dot above the 6. Running total: 3.
- Final: write **3** under the column.
- Count dots: **2**. Carry 2 to the next column.

. .

5728
3417
6892
4536+
3

Continue for tens, hundreds, thousands columns. Reveal answer.

- “Notice: *I never stopped to think ‘what’s $8 + 7 + 2 + 6$?’ I just scanned for pairs that make 10.*”

Activity (15 min) — First try

- Students re-do the same problem using the dot method on the *back* of their baseline sheet.
- Time themselves again.
- Compare: how much slower (or faster) was the dot method on the FIRST try?

Wrap-up (5 min)

- Show of hands: who was faster with the dot method on first try? Who was slower? (Most will be slower — that’s expected. The speedup comes after 3–4 days of practice.)

- Preview Day 2: “Tomorrow we’ll work on the foundational skill — instantly seeing pairs that make 10.”

Student-facing content

The Dot Method (*Bindu Paddhati*)

Instead of adding digits one by one, **scan** the column for pairs that make 10. Mark a dot above the digit that “completes” each 10. The number of dots = your carry.

Why? Adding 10 is easy. Adding 9 to 4 to 7 to 6 takes thought. If you can group into 10s first, you only count the leftover and the dots.

Today’s baseline: my time using my normal method = ____ seconds. My time using dot method (first try) = ____ seconds. I expect this gap to shrink by Day 5.

Homework

- Practice the “make 10” reflex: 10 quick pairs. $8+?$ = 10. $3+?$ = 10. $6+?$ = 10. $4+?$ = 10. $9+?$ = 10. $1+?$ = 10. $7+?$ = 10. $5+?$ = 10. $2+?$ = 10. $0+?$ = 10.
- Do all 10 in under 15 seconds.

Differentiation

- **Tier up:** Try the dot method on a 5-addend $\times$ 5-digit problem. Time it.
- **Tier down:** If 4-digit columns are too hard, drop to 3-addend $\times$ 3-digit for today. We’ll scale up.

Teacher notes

- The baseline time is the *single most important* artefact of Day 1. Make sure every student records it honestly. Day 5’s retest depends on it.
- Don’t promise a specific speedup — say “you’ll likely be faster by Day 5; how much depends on you.”
- Walk around during the demo and notice who looks confused. Pull them aside in the wrap-up minute.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf*, §5.4 — Columns of Figures (Vedic Cultural Center, Sammamish WA)

Day 2 — Making Tens

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will instantly recall the 10-pair of any single digit (0+10, 1+9, 2+8, 3+7, 4+6, 5+5) AND identify all 10-pairs in a list of 6–8 single-digit numbers in under 10 seconds.

Materials

- A class wall-poster of the 5 pairs (1+9, 2+8, 3+7, 4+6, 5+5). Display before class.
- Pre-printed pair-drill sheets (lists of 8 digits each — students circle 10-pairs).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Make-10 lightning round.** Teacher calls 10 single digits in 30 seconds; students chorus the partner. (8 → 2. 3 → 7. 6 → 4. ...)
- Repeat with a different sequence.

Core (15 min) — The 10-pair pattern

1. **Reveal the wall poster.** Walk through the 5 pairs.
2. **Note (write on board):** “Every single digit from 1–9 has exactly ONE partner that makes 10. Knowing all 5 pairs gives you the whole table.”
3. **Mental check:** call out digits randomly for 60 seconds; class chorus the partner.
4. **The “extra” patterns:** if the digit pair sums to 11 instead of 10, write down 1 and “the extra” goes into running total. If sum is 9 (no 10-pair available), just count.

Example column: 7, 5, 3, 8

- Scan: $7 + 3 = 10$. Mark dot above 3. Running = 0.
- 5. Running = 5.
- 8. Running = 13 → 10-pair found (3 already used). Mark dot above 8. Running = 3.
- Write 3 under column. Carry 2 dots.

Activity (20 min) — Pair drills

- Hand out drill sheets. Each sheet has 10 rows, each row a list of 6–8 digits.
- Working in pairs:
 - Partner A reads a row aloud.
 - Partner B identifies all 10-pairs in the row in under 10 sec.
 - Swap roles for the next row.

Suggested rows:

Row 1: 3 7 2 5 8 1 4 6
Row 2: 9 4 1 2 8 5 3 7
Row 3: 5 5 6 4 1 9 2 8
Row 4: 7 3 6 1 9 8 2 4
...

After 5 minutes of paired drills, do a class-level **timing test**: project a single row; students individually find all 10-pairs and write count + leftover. 20 seconds per row. Do 5 rows.

Wrap-up (5 min)

- Quick check: “Without thinking, what makes 10 with: 6?” (chorus 4). “With 8?” (chorus 2).
- Preview Day 3: “Now we’ll add a *single column* of numbers using this.”

Student-facing content

The Five 10-Pairs

$1 + 9$

$2 + 8$

$3 + 7$

$4 + 6$

$5 + 5$

Practice until you can name the partner of any single digit *instantly* — no thinking. This is the foundational skill of the dot method.

Drill: in a row of 6–8 digits, scan and circle every 10-pair you can find. Write what’s “left over.”

Example: in 3 7 2 5 8 1 4 6, you can find: 3+7, 2+8, 4+6 — three pairs, leftover is 5 1.

Homework

- Practice the 10-pair recall. Have a parent or sibling call out single digits; you answer the partner. Beat 20 digits in 30 seconds.
- Continue: in any list of 6 digits, find all 10-pairs in under 10 sec.

Differentiation

- **Tier up:** Find pairs that sum to 9 OR 10 OR 11. (For future Day 7 — numbers near a base.)
- **Tier down:** Stick to recall-only practice today. The column work can wait one day.

Teacher notes

- This is the most important fluency lesson of the module. Do NOT skip the lightning round.
- If a student can’t recall pairs by end of class, assign extra drill at home and provide a “pairs cheat sheet” for the rest of the module — withdraw it on Day 7.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf*, §5.2 Mental Addition.

Day 3 — Single Column with Dots

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will add a single column of 4–6 single-digit numbers using the dot method, correctly identifying both the units digit and the number of dots (the carry).

Materials

- Worksheet: 10 single columns of 4–6 digits each (no carrying yet — just write the result + count dots).
- Whiteboard.

Lesson flow

Warm-up (5 min)

- Make-10 lightning round (30 sec).
- Recall: “*In any list of digits, how do I find the units digit if I’m using the dot method?*” (Take a student answer. Reinforce: scan, mark dots, the unwritten leftover is the units.)

Core (15 min) — The procedure formalised

Write on the board, large:

THE DOT METHOD FOR ONE COLUMN

1. Start with running total = 0.
2. Scan top to bottom.
3. When adding a digit makes total ≥ 10 , mark a DOT above that digit. Subtract 10 from the running total.
4. Continue.
5. At the end:
 - Final running total is the UNITS DIGIT (write below).
 - Number of dots = CARRY (to next column).

Worked example on the board:

```
6
3
7 ← total 6 → 9 → 16 ≥ 10, dot ·, total = 6
5
8 ← total 6 → 11 ≥ 10, dot ·, total = 1
4
2
—
1 ← running total at end
... wait, let me recount
```

(Be honest with yourself — recount aloud. Models the self-checking habit.)

Correct walk-through: - Start: 0 - + 6 → 6 - + 3 → 9 - + 7 → 16. Total ≥ 10 : mark · above 7. Total = 6. - + 5 → 11. Total ≥ 10 : mark · above 5. Total = 1. - + 8 → 9. - + 4 → 13. Total ≥ 10 : mark · above 4. Total = 3. - + 2 → 5. - **Units: 5. Dots: 3. Carry 3 to next column.**

Verify by standard addition: $6+3+7+5+8+4+2 = 35$. Units = 5. Carry = 3. ✓

Activity (20 min) — Single-column drill

- Hand out worksheet of 10 columns.
- 5 minutes individual work.
- 10 minutes paired correction: partner-A solves column 1 aloud while partner-B watches and corrects. Swap. Continue alternating.
- Final 5 minutes: silent retry on any column students got wrong.

Wrap-up (5 min)

- Hand out the **formative quiz** (quizzes/formative.md Part A) — 4 quick column problems, 3 min.
- Collect; mark overnight.
- Preview Day 4: “Now we put multiple columns together — with the carries flowing from one to the next.”

Student-facing content

Dot method, one column:

1. Running total = 0.
2. Scan top to bottom.
3. When total ≥ 10 : mark a dot, subtract 10.
4. Final running total = units digit. Dot count = carry.

Example: column 6, 3, 7, 5, 8, 4, 2

6
3
7
5
8
4
2

5

← units

3 dots = carry 3 to the next column

Homework

- Solve the back of the worksheet (10 more single-column problems). Self-check answers against standard addition.

Differentiation

- **Tier up:** Solve columns of 8–10 numbers each (challenge sheet).
- **Tier down:** Stick to columns of 4 numbers each. Build up.

Teacher notes

- Common error: students forget that the dot count IS the carry — they recount the dots wrongly. Reinforce: 1 dot = +1 carry, 2 dots = +2 carry, etc.

- A second common error: students mark a dot but forget to subtract 10 from the running total — so they keep counting up. Watch for total > 20 mid-column; if so, they missed a subtraction.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf*, §5.4 Columns of Figures (the formal procedure).

Day 4 — Multiple Columns

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will add a 4-addend $\times$ 4-digit column-sum using the dot method, correctly propagating carries between columns.

Materials

- Day 3 formative quiz returned with marks.
- Worksheet: 8 multi-column problems (varying from 3×3 -digit to 5×4 -digit).
- Whiteboard / large grid paper.
- Stopwatch.

Lesson flow

Warm-up (5 min)

- Make-10 lightning round.
- Hand back yesterday's formative. Don't review — students see their marks privately.

Core (10 min) — The carry handoff

- Demonstrate on the board:

```

5728
3417
6892
4536+

```

Units column: $8 + 7 + 2 + 6$.

- $0 \rightarrow 8 \rightarrow 15$ (dot on 7, total 5) $\rightarrow 7 \rightarrow 13$ (dot on 6, total 3) $\rightarrow 3$.
- **Write 3. Carry 2.**

. .

```

5728
3417
6892
4536+
  3 [carry 2]

```

Tens column: $2 + 1 + 9 + 3 + \text{carry-}2$.

- Include the carry IN the column (write it as a small digit at the top of the column, or hold it in your head).
- $0 \rightarrow 2 \rightarrow 3 \rightarrow 12$ (dot on 9, total 2) $\rightarrow 5 \rightarrow 7$ (the carry).
- **Write 7. Carry 1.**

Continue.

Final result: **20573**

- Verify with calculator or standard algorithm.

Activity (25 min) — Drill

Follow `activities/activity-01-column-drill.md`. Brief:

- 8 multi-column problems, individual work first (8 min).
- Then paired check (10 min): partners trade sheets; one reads problem aloud while the other re-solves on a fresh sheet; compare results.
- Final 7 min: any 3 problems the student got wrong, redo silently.

Wrap-up (5 min)

- Show of hands: who finished all 8 in 8 minutes? Who needs more time? (Don't shame either group — Day 5 is the retest.)
- Preview Day 5: timed re-test of Day 1's baseline problem.

Student-facing content

The dot method with carries

When you have multiple columns, the dots in column N become the *carry* added to column N+1.

Add the carry IN the column as you scan — either by including it as a tiny digit at the top of the column, or by adding it to your starting running total.

Practice this until you can do a 4×4-digit sum in under 60 seconds.

Homework

- Find a column-addition problem in any textbook. Solve it BOTH ways (standard + dot method). Time yourself for each. Bring tomorrow.

Differentiation

- **Tier up:** 5×5-digit sums; aim for under 90 seconds.
- **Tier down:** 3×3-digit sums until comfortable; scale up.

Teacher notes

- The carry handoff is where most errors happen. Two common patterns:
 1. Forgetting to include the carry in the next column.

- 2. Including it twice (once as the carry, once as part of the running total).
- For each, slow down with the student and have them write the carry IN the column as a small digit before they start scanning.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf*, §5.4 (the carry handoff example on page 19).

Day 5 — Mid-Module Check (Speed Retest)

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will retest their Day 1 baseline using the dot method and submit a project plan paragraph.

Materials

- Each student’s Day 1 baseline sheet (returned to them).
- Fresh copy of the SAME problem ($5728 + 3417 + 6892 + 4536$).
- Formative quiz ([quizzes/formative.md](#) Part B) — 5 mixed problems.
- Project brief handouts ([assessments/project-brief.md](#)).

Lesson flow

Warm-up (3 min)

- Make-10 lightning round (30 sec; should be effortless by now).
- Brief: “Today is your retest. The same problem you did on Day 1, with the new method. We’ll see how much you’ve sped up.”

Speed retest (5 min)

- Hand out the fresh sheet.
- On “Go,” solve using dot method. Record time.
- Compare to Day 1 baseline. Average gain: target 50% reduction in time.

Formative quiz (15 min) — [quizzes/formative.md](#) Part B

- 5 mixed problems:
 - 2 single-column adds
 - 2 multi-column 3×3 adds
 - 1 multi-column 4×4 add (the retest doesn’t count)
- Closed notebook.
- 12 min to solve + 3 min for class review.

Project commitment (15 min)

- Hand out project brief.
- Read aloud the three project formats:
 1. **60-second teaching video** — teach a younger student (Grade 4) the dot method.
 2. **Printed workbook** — 20 problems with worked solutions using the dot method, designed for self-study.
 3. **Comparative speed-test data project** — collect Day-1-style baseline + Day-5-style retest data from 5–10 family members or schoolmates. Plot the results.
- Pairs (or individuals) choose a format AND a partner by end of class.
- Each pair writes a 1-sentence “what we’ll do” on a sticky note, posts to the board under their format.

Wrap-up (7 min)

- Show of hands by gain category: <0% (slower), 0–25% faster, 25–50% faster, 50%+ faster.
- Note: *“Whatever your gain, you’ve learned a real technique that will keep speeding up with use.”*
- Preview Day 6: *“Tomorrow we ask: why does this work? You’ll write a paragraph explaining it in your own words.”*

Student-facing content

The retest

Same problem as Day 1: $5728 + 3417 + 6892 + 4536 = ?$

Day 1 time (baseline): ____ sec Day 5 time (dot method): ____ sec % **change**:

Project choice (decide today):

1. 60-second teaching video
2. Printed workbook (20 worked problems)
3. Comparative speed-test data project

Partner: _____ Format: _____

One sentence: what we’ll do —

Homework

- Submit project plan paragraph (one paragraph, 4–6 sentences) by tomorrow morning.
- Begin gathering materials.

Differentiation

- **Tier up:** If you got 60%+ speedup, your project should challenge you. Pick the speed-test data project — it requires statistical thinking.

- **Tier down:** Pick the teaching video — the act of *teaching* the method reinforces it for you.

Teacher notes

- Be honest with students whose times slowed: the method has a learning curve, and steady drill (Day 7’s mental patterns will help) will catch them up.
- Mark the formative quiz tonight.
- Speed-test data projects need ethics: pairs should ask permission before timing family members.

Citation(s) used in this lesson

- Source comparison: see `sources.md`.

Day 6 — Why It Works

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will explain in their own words why the dot method gives the same answer as the standard algorithm, citing place value and the fact that a “dot” represents a 10 carried.

Materials

- Whiteboard.
- The 16-Sūtra list (Microsoft Word - sutras1.pdf) — printed and visible.
- Worksheet: “explain to a friend” prompt + space.

Lesson flow

Warm-up (5 min)

- Make-10 round.
- Hand back formative. Discuss any common error pattern from the quiz briefly.

Core (20 min) — The mathematical reasoning

1. **The core claim** (write boxed on the board):

A dot in a column represents “10 has been used up.” Since 10 in any column equals 1 in the next column, the dot count is the carry.

2. **Walk through a simple example to prove it:**

Column: 8, 7, 2, 6, 4

Direct addition: $8+7+2+6+4 = 27$.

Units = 7. Carry = 2.

Dot method:

Start 0. $+8 = 8$. $+7 = 15 \rightarrow$ dot $\cdot$ on 7, total = 5.
 $+2 = 7$. $+6 = 13 \rightarrow$ dot $\cdot$ on 6, total = 3. $+4 = 7$.
Units = 7. Dots = 2. Carry = 2.

Same answer.

3. **Why? Because each dot is precisely “I just crossed 10.”**

- Crossing 10 once = 1 carry.
- Crossing 10 twice = 2 carry.
- The leftover is just whatever remains in the running total.

4. **Connection to Tirthaji’s 16 sūtras.** Display the list:

- Sūtra #7: **Sankalana Vyavakalanābhyām** — “by addition and by subtraction.”
- Sūtra #8: **Pūraṇāpūraṇābhyām** — “by completing or non-completing.”
- The dot method is *not* itself one of these sūtras — it’s a 20th-century pedagogical technique that uses the *spirit* of these sūtras (looking for completions to a base).
- Be honest: “There is no ancient Sanskrit verse that says ‘use the dot method.’ What we have is the *idea* of completion that classical Indian math used, and a modern teaching tool that makes it concrete.”

Activity (15 min) — Explain to a friend

- Worksheet prompt: > Imagine your Grade 4 cousin asks: “*Why does that dot trick work? Isn’t it just magic?*” Write 4–6 sentences explaining why it’s NOT magic. Use the words *carry*, *place value*, and *10*.
- 10 min individual writing. 5 min pair-sharing: read your paragraph to your partner; they tell you one part that’s clear and one part that’s unclear.

Wrap-up (5 min)

- Read aloud one strong paragraph (with permission).
- Preview Day 7: “*Tomorrow — using the same ‘completion to 10’ trick for mental addition of 2-digit numbers near a base.*”

Student-facing content

Why the dot method works

A column of digits is in some place-value column — units, tens, hundreds. When the running total reaches 10, that’s literally **one** of the next place value. So the dot is just a *visible* carry.

The 16 sūtras of Vedic mathematics (Tirthaji, 1965) include *Sankalana Vyavakalanābhyām* (“by addition and by subtraction”) and *Pūraṇāpūraṇābhyām* (“by completing or non-completing”) — both use the *idea* of looking for completions to a base (like 10). The dot method makes this idea concrete and visible.

It is not magic. It is good notation.

Homework

- Find ONE example in your daily life where you naturally “complete to 10” without thinking. (E.g. adding tip to a bill — round to 10. Counting change. Adding minutes to a clock.) Write 2 sentences describing it.

Differentiation

- Tier up:** Write a 1-paragraph explanation suitable for a Grade 11 student. Include the connection to base-10 representation and to the carry algorithm.
- Tier down:** Skip the writing; just verbalise the explanation to your partner.

Teacher notes

- The honest “not in any sūtra” framing matters. Some students (or parents) believe the dot method is “ancient Vedic mathematics” — clarify gently. The achievement here is good *teaching*, which is also a real achievement.
- The most common error in student paragraphs: confusing the dot with “skip counting.” Reinforce that the dot represents *the carry*, not “I’m skipping a number.”

Citation(s) used in this lesson

- Microsoft Word - sutras1.pdf* — list of 16 Tirthaji sūtras (sūtras #7 and #8 specifically).
- Optional teacher reference: *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, full treatment.

Day 7 — Mental Math Patterns: Numbers Near a Base

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will add 2-digit numbers ending in 9, 19, 29, ... (or 8, 18, 28, ...) mentally, by adding the round base and subtracting the deficiency.

Materials

- Whiteboard.
- Drill sheet for Activity 02 ([activities/activity-02-near-base.md](#)).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- Make-10 round.
- Quick recall: “*In yesterday’s writing, what was the most surprising place where you ‘complete to 10’ without thinking?*” — 2 student shares.

Core (15 min) — The “near-base” pattern

1. Write on board:

$$33 + 9 = ?$$

Most students will scratch out $33+9 = 42$ in their head, taking 2–3 seconds.

2. Show the trick:

$$9 = 10 - 1$$

$$\text{So } 33 + 9 = 33 + 10 - 1 = 43 - 1 = 42$$

Now do it in your head: **add 10, subtract 1**. ~0.5 seconds.

3. **The general pattern (write on board):**

To add a number that ends in 9, 8, 7 (close to a multiple of 10):

1. ROUND UP to the multiple of 10.
 2. ADD that.
 3. SUBTRACT the deficiency.
4. Worked examples (class chorus):

- $47 + 9 = 47 + 10 - 1 = 56$

- $28 + 19 = 28 + 20 - 1 = 47$

- $64 + 29 = 64 + 30 - 1 = 93$

- $55 + 48 = 55 + 50 - 2 = 103$

- $36 + 38 = 36 + 40 - 2 = 74$

5. **Why?** Same “completion to a base” idea — and now the base can be 10, 20, 30, 100. This is exactly Tirthaji’s sūtra #7 (*Sankalana Vyavakalanābhyām* — “by addition and by subtraction”). Write the name on the board.

Activity (20 min) — Near-base drills

Follow `activities/activity-02-near-base.md`. Brief:

- 5 min individual: 20 problems on a sheet, do as many as you can.
- 5 min paired: partner-A reads a problem aloud, partner-B answers mentally in <3 seconds, swap.
- 5 min mixed: 10 problems on the board; class shouts answer in chorus.
- 5 min quick-fire: teacher calls a 2-digit number ending in 9 or 8, then any other 2-digit number; students whisper-answer to their neighbour.

Wrap-up (5 min)

- Show of hands: who can now add + 9 to any 2-digit number in under 1 second? (Should be most of the class.)
- Preview Day 8: project work begins.

Student-facing content

Add a number near a base (10, 20, 30, 100)

1. Round it UP to the base.
2. Add that.
3. Subtract the deficiency.

Examples: $- 33 + 9 = 33 + 10 - 1 = 42$ - $47 + 19 = 47 + 20 - 1 = 66$ - $55 + 48 = 55 + 50 - 2 = 103$ - $234 + 99 = 234 + 100 - 1 = 333$

This pattern is named in Tirthaji's 7th sūtra: ***Sankalana Vyavakalanābhyām*** — “by addition and by subtraction.”

Homework

- 20 mental additions. Do them in under 60 seconds. List on the back of today's worksheet.
- Bring all project materials tomorrow.

Differentiation

- **Tier up:** Apply the same idea to numbers near 100 (e.g. $234 + 98 = 234 + 100 - 2$). Also try 3-digit additions in your head.
- **Tier down:** Stick to “+9” first. Once that's automatic, add “+19, +29, +39.” Build up.

Teacher notes

- Some students will object: “But subtracting is HARDER than adding.” Not really — subtracting 1 or 2 is trivial once you've done the bigger jump. Push through the initial resistance with the drills.
- This lesson is also a preview of Module 9 (Subtraction by Complements). Don't elaborate now; just note that *Nikhilam* (sūtra #2) and other complement techniques are coming.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf*, §5.5 By Addition and By Subtraction (page 19–20).
- *Microsoft Word - sutras1.pdf* — sūtra #7 Sankalana Vyavakalanābhyām.

Day 8 — Project Session 1: Plan and Begin

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, every pair will have a written project plan and at least one tangible artefact (script outline, first 5 workbook problems drafted, or first 3 data points collected) for their final project.

Materials

- All project materials brought by students.

- A printed rubric (assessments/rubric.md) per pair.
- Stopwatch.

Lesson flow

Warm-up (5 min)

- Make-10 round.
- Hand out the rubric. 60-second walk-through.

Core (35 min) — Workshop sprints

Same three-sprint structure as Panchabhuta Day 8:

Sprint 1 (10 min) — Plan completion. Every pair finalises: - Format (video / workbook / data project) - One-sentence audience takeaway - Three-part structure of the artefact - Division of labour - The PDF section they'll cite (e.g. "Vedic Mathematics Batch1.pdf, §5.4")

Sprint 2 (10 min) — First artefact. - **Video projects:** draft script in 10 min. Rough storyboard. No filming yet. - **Workbook projects:** draft 5 problems with worked dot-method solutions. - **Data projects:** collect baseline times from 3 family members or schoolmates (in school: use classmates). Record in a table.

Sprint 3 (10 min) — Build / iterate. Continue.

Teacher's role: - 60-sec rotation per pair in Sprint 2. - Force a "stop planning, start producing" intervention for any pair still drafting their plan in Sprint 2. - For video projects: confirm by Sprint 3 that they have a clear shot list, not just a script. - For data projects: confirm by Sprint 3 that they have at least 3 real data points.

Wrap-up (5 min)

- 30-sec share-out from 4 pairs (you call them).
- Plan paragraph submitted at the door.
- Preview Day 9: refine and rehearse.

Student-facing content

Project workshop — Session 1

By end of class, your pair should have:

1. A written project plan paragraph.
2. The start of your artefact: a script draft, 5 worked problems, OR 3 data points.

Plan template:

Format: _____ (video / workbook / data project)

Audience walks away thinking: _____

Structure (3 parts):

1. _____
2. _____
3. _____

Division of labour (Partner A): _____
Division of labour (Partner B): _____
Source we'll cite: _____ (PDF + section)

Homework

- Whatever didn't get done in class — finish at home tonight.

Differentiation

- **Tier up:** Stretch goal — add a comparison element. (E.g. video: include a side-by-side speed comparison; workbook: include 5 standard-algorithm problems for contrast; data: also test mental “near-base” additions.)
- **Tier down:** Allow simpler format mid-stream. A clean workbook beats a half-built video.

Teacher notes

- For video projects: students often shoot poor-quality video at home. Encourage them to rehearse and shoot in one or two takes — the *teaching content* matters more than production value.
- For data projects: be wary of small-sample misinterpretation. If a pair has only 3 data points and one shows the dot method is slower, that's a real finding — let them report it. Statistical thinking is part of the project.

Citation(s) used in this lesson

- Pair-chosen source from `sources.md`, verified during Sprint 1.

Day 9 — Project Session 2: Refine and Rehearse

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, every pair will have a presentation-ready project artefact and will have rehearsed their 3-minute presentation at least once with timed feedback.

Materials

- All in-progress materials.
- Stopwatch.
- Feedback slip per pair: 1 strength, 1 fix.

Lesson flow

Warm-up (5 min)

- Make-10 round.

- Brief: “Today is refine + rehearse. By end of class, you’ll have presented to one other pair and gotten feedback.”

Core (30 min) — Two halves

Half 1 (15 min) — Refine. - Teacher rapid rotation: 60 sec per pair, focused question: “What’s your strongest 30 seconds?” - Watch for: - Pairs without a clear demonstration of the method (essential for video and workbook). - Workbook projects with errors in the worked solutions — flag and fix. - Data projects with ambiguous tables — make them annotate. - Pairs that left planning incomplete in Day 8 — push them to finish now. - Pairs that finish early start rehearsing.

Half 2 (15 min) — Rehearse with a buddy pair. - Pair A presents to Pair B for 3 minutes (use the timer). - Pair B gives feedback: 1 strength + 1 fix, 60 seconds. - Swap. Total ~8 min per rotation $\times 2 = 16$ min.

Wrap-up (5 min)

- One-sentence share: “What’s the one thing I’m changing before tomorrow?”
- Tomorrow: presentations + summative quiz.

Student-facing content

Project workshop — Session 2: Refine + Rehearse

Today’s checklist: - ☐ My artefact is presentation-ready. - ☐ I cited at least one source (PDF + section). - ☐ My demonstration of the method is clear. - ☐ I’ve timed my presentation — it fits in 3 minutes. - ☐ I know what my partner says. - ☐ I’ve heard feedback from one other pair and decided what to change.

Tomorrow you present.

Homework

- Final touches at home. Bring everything tomorrow.

Differentiation

- **Tier up:** Anticipate one tricky audience question and prepare a 30-sec answer (e.g. “*Why is dot method ‘better’ than standard?*” — honest answer: not better in all cases; better for multi-addend columns).
- **Tier down:** Allow reading from notes during presentation. Goal is communication.

Teacher notes

- For video projects: if a pair hasn’t yet filmed, push them HARD to film in the second half. Don’t allow “we’ll film at home tonight” — too often that becomes “we didn’t have time.”
- For data projects: check that the plot or table is clearly labelled.

Citation(s) used in this lesson

- Pair source verified per pair.

Day 10 — Final Assessment + Project Showcase

Module: Bindu Paddhati · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate mastery of the dot method via a summative quiz and a 3-minute project presentation.

Materials

- Summative quiz (quizzes/summative.md).
- Stopwatch.
- Rubric per pair.
- Reflection slip per student.

Lesson flow

Warm-up (2 min)

- Brief: “Quiz first (15 min), presentations after.”

Part 1: Summative Quiz (15 min)

- Distribute quizzes/summative.md. Closed notebook.
- Quiz covers: make-10 fluency (timed), single column with dots, multi-column with carry, near-base mental addition, one short-answer “why does it work.”
- Collect at 15-min mark.

Part 2: Project Presentations (25 min)

- 3 min presentation + 1 min Q&A per pair (4 min total).
- ~6 pairs fit in 25 min. **Reserve next class for remaining pairs** if needed.
- During each:
 - Teacher marks rubric in real-time.
 - One assigned audience member asks one question.
 - Other audience members fill “I learned / I wonder” slips.

Wrap-up (3 min)

- Closing question (oral): “*What’s one mental-math habit you’ll keep using after this module?*”
- Reflection slip submitted.

Student-facing content

Today: quiz + presentations

15 min for the summative quiz (closed book).

3-minute presentation per pair, then 1-min audience question.

Reflection (due at end of class): 1. What did you set out to make/learn, and what did you actually end up with? 2. What’s the most surprising thing you discovered while making this? 3. If you had three more days, what would you do differently?

Homework

- None. Module is complete.

Differentiation

- **Tier up:** Students finishing the quiz early can write a 1-paragraph “what would Module 5 look like for Grade 4?” — input for Primary band.
- **Tier down:** Open-notebook quiz for students with documented learning needs.

Teacher notes

- Mark quiz immediately after class while presentations are fresh.
- If presentations spill into Day 11, protect that time.

Citation(s) used in this lesson

- All sources verified.

Module 5 (Middle) — Formative Quizzes

Two parts: Part A on Day 3, Part B on Day 5.

Part A — Day 3 (3 min, 4 single-column problems)

Name: _____ **Class:** _____

Solve each column using the dot method. Show your dots. Write the units digit AND the carry.

1.

4
7
6
3
2

Units: ____ Carry (dots): ____

2.

5
8
1
9
6

Units: ____ Carry: ____

3.

2
4
9
8
7
3

Units: ____ Carry: ____

4.

6
3
5
7
4
8
1

Units: ____ Carry: ____

Part A answer key

Q	Sum	Units	Carry	Common error
1	22	2	2	Missing dot above 7 ($4+6 = 10$ also valid; multiple correct dot placements)
2	29	9	2	Missing the $1+9 = 10$
3	33	3	3	Three dots — students often write 2
4	34	4	3	Three dots; common to miss $6+4 = 10$

Part B — Day 5 (12 min, 5 mixed problems)

Name: _____ Class: _____

1. (2 marks) Single column with dots.

3
8
9
4
6
2

Units: ____ Carry: ____

2. (3 marks) Multi-column addition with dot method.

247
395
178
+

Answer: _____

3. (3 marks) Multi-column addition with dot method.

5683
2917
4256
1845+

Answer: _____

4. (1 mark each, mental) Near-base addition.

- a) $47 + 9 =$ _____
- b) $64 + 19 =$ _____
- c) $38 + 29 =$ _____
- d) $55 + 48 =$ _____

5. (2 marks, short answer) Why does adding a dot work? Explain in one sentence.

Part B answer key

Q	Answer	Marks
1	32; units 2, carry 3	2 (1 each)
2	820	3 (1 per column correct)
3	14701	3 (1 per row of 2 columns correct)
4a	56	1
4b	83	1
4c	67	1
4d	103	1
5	A dot represents 10 in the column, which equals 1 in the next column — so counting the dots IS the carry. (Variants of this acceptable.)	2 if it mentions both 10 and carry; 1 if only one.

Total Part B: 14 marks.

What to do with the results

- Class average below 8/14 = retreach Day 6 on “why it works”; reinforce make-10 fluency in warm-ups.
- Individual below 5/14 = pull aside on Day 7 for 10-min reteaching of multi-column carries.

Module 5 (Middle) — Summative Quiz

Time: 15 min · **Marks:** 25 · **Closed notebook**

Instructions: Name + class at top. Show your dots on column-addition problems. No calculators.

Name: _____ **Class:** _____

Section A — Make-10 Fluency ($4 \times 0.5 = 2$ marks, 1 min)

Write the partner that makes 10:

8 + ____ 6 + ____ 3 + ____ 9 + ____

Section B — Single Column ($2 \times 2 = 4$ marks)

Use the dot method. Show dots. Write units AND carry.

1.

5
3
7
8
2
4
6

Units: ____ Carry: ____

2.

9
2
4
8
5
6

1
7

Units: ____ Carry: ____

Section C — Multi-column Addition ($3 \times 3 = 9$ marks)

Use the dot method. Show dots in each column.

3.

436
582
197
+

Answer: _____

4.

2849
3576
4928
+

Answer: _____

5.

58273
14916
62385
27548
+

Answer: _____

Section D — Mental Near-Base Addition ($5 \times 1 = 5$ marks)

Do these in your head. Write only the answer.

- a) $36 + 9 =$ _____
 - b) $47 + 19 =$ _____
 - c) $83 + 29 =$ _____
 - d) $145 + 39 =$ _____
 - e) $246 + 98 =$ _____
-

Section E — Open Response (5 marks)

6. Imagine a Grade 4 student says: “*The dot method is just magic. Why does it work?*” In 4–5 sentences, explain why it’s not magic. Use the words **carry**, **10**, and **place value**.

Answer key

Q	Answer	Marks
Section A	2, 4, 7, 1	2 (0.5 each)
1	sum 35; units 5, carry 3	2
2	sum 42; units 2, carry 4	2
3	1215	3 (1 per column)
4	11353	3
5	163122	3
6a	45	1
6b	66	1
6c	112	1
6d	184	1
6e	344	1
6 (open)	4–5 sentence explanation referencing carry, 10 as base, place value.	5: 2 for correct reasoning; 2 for using the three required words; 1 for clarity.

Total: 25 marks.

Grading bands (project combined → out of 40): - 32–40: Exceeds - 24–31: Meets - 16–23: Approaches - 0–15: Beginning

Homework

Homework — Module 5 Dot Addition (Middle)

Day 1 → Day 2

Task: Practice the 10-pair reflex. Do all 10 in under 15 seconds: $8+?=10$ $3+?=10$ $6+?=10$ $4+?=10$ $9+?=10$ $1+?=10$ $7+?=10$ $5+?=10$ $2+?=10$ $0+?=10$

Time: 5 min.

Day 2 → Day 3

Task: Have a parent or sibling call out 20 single digits in 30 seconds; you say the 10-partner for each. Track how many you got right.

Time: 5 min.

Day 3 → Day 4

Task: Solve the 10 column problems on the back of today's worksheet using the dot method. Self-check against standard addition.

Time: 15 min.

Day 4 → Day 5

Task: Find any column-addition problem in any textbook. Solve it BOTH ways (standard + dot method). Time yourself for each. Bring tomorrow.

Time: 10 min.

Day 5 → Day 6

Task: Submit your project plan paragraph in writing (4–6 sentences).

Time: 15 min.

Day 6 → Day 7

Task: Find ONE example in your daily life where you naturally “complete to 10” without thinking (rounding a bill, counting change, adding minutes to a clock). Write 2 sentences.

Time: 10 min.

Day 7 → Day 8

Task: 20 mental near-base additions. Do them in under 60 seconds. List on the back of today's worksheet. Bring ALL project materials tomorrow.

Time: 5 min (plus material gathering).

Day 8 → Day 9

Task: Finish at home what you didn't finish in class. Tomorrow is for refining, not starting.

Time: capped at 30 min.

Day 9 → Day 10

Task: Final touches. Make sure you have: the artefact, your script/notes, your 3-min presentation timed in your head.

Time: 15 min.

Writing Prompts

Writing Prompts — Module 5 Dot Addition (Middle)

Six prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

- 1. (Day 1, end of class, 3 min)** What does it feel like to add a column of numbers under time pressure? Describe what your mind does — do you talk through the addition, see numbers, count on fingers?
- 2. (Day 7, end of class, 5 min)** Yesterday you had to write “why the dot method works.” Today you used it for mental addition. Has your *feeling* about math speed changed in the last week? Be honest — even if the answer is “no.”

Explanatory prompts

- 3. (Day 3 or Day 6, homework, ~120 words)** Explain the dot method to a Grade 4 student who has never seen it. No more than 6 sentences. Use one specific example.
- 4. (Day 7, homework, ~100 words)** Explain the *Sankalana Vyavakalanābhyām* idea (add to the round number, take away the difference) in your own words. Use one example with addition and one with subtraction.

Argumentative prompt

- 5. (Day 6 or homework, 1 paragraph)** Some people claim the dot method is “ancient Vedic mathematics.” Others say it’s a 20th-century pedagogical technique inspired by Vedic mathematics. Pick a position and defend it in 5–7 sentences. Use what you know about the 16 sūtras.

Creative prompt

6. (Day 4 or Day 5, optional, 1 page) Write a short script (50–150 words) in which a frustrated student and a confident dot-method user have a conversation. The frustrated student raises one good objection to the dot method. The confident user gives a clear, honest answer. (No “and then they magically agreed” endings — let the disagreement be real.)

Prompt-to-band notes

- For **Primary** band, prompts 1, 3, and 6 work as drawings or oral responses.
- For **Senior** band, prompt 5 deserves a full essay treatment with citations from `sources.md`.

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction.
- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the rubric — does the writing acknowledge the other side, give a specific example, reach a clear position?
- Prompt 6: no grading. Read the best ones aloud (with permission).

Activity 01 — Column Drill (Multi-column)

Module: Bindu Paddhati · **Band:** Middle · **Day:** 4 · **Time:** 25 min

Type

Paired practice with self-correction

Goal

Each student solves 8 multi-column addition problems using the dot method, with the partner checking accuracy.

Materials per pair

- 1 printed problem sheet (8 problems, see below)
- 1 stopwatch
- 1 pen each

Set-up (teacher, before class)

- Print the problem sheet (you can vary the numbers — keep difficulty graded from 3×3-digit to 5×4-digit).

Instructions (student-facing handout)

Multi-column Drill — work in pairs.

1. Both partners solve all 8 problems individually (8 min). Don't look at each other's work.
2. Now swap sheets. Partner-A reads problem 1 aloud; partner-B re-solves it on a fresh sheet (no peeking at partner-A's answer). Compare answers. If they differ, work out which is correct.
3. Repeat for problems 2–8, alternating who reads.
4. Last 5 min: each student re-does any 3 problems they got wrong, silently.

Problem sheet

1. $\begin{array}{r} 234 \\ 356 \\ + 198 \end{array}$	2. $\begin{array}{r} 587 \\ 342 \\ + 925 \end{array}$	3. $\begin{array}{r} 429 \\ 718 \\ + 386 \end{array}$	4. $\begin{array}{r} 163 \\ 274 \\ 395 \\ + 822 \end{array}$
5. $\begin{array}{r} 2847 \\ 3576 \\ 4928 \\ + 1235 \end{array}$	6. $\begin{array}{r} 5683 \\ 2917 \\ 4256 \\ + 1845 \end{array}$	7. $\begin{array}{r} 38271 \\ 45982 \\ 36475 \\ 29384 \\ + 41728 \end{array}$	8. $\begin{array}{r} 58273 \\ 14916 \\ 62385 \\ + 27548 \end{array}$

[Answers (teacher copy):

1. 788 2. 1854 3. 1533 4. 1654
5. 12586 6. 14701 7. 191840 8. 163122]

Discussion prompts (last 2 min)

- Which problem took the longest? Why?
- Did your partner's solution differ from yours on any problem? Whose was right?
- Were the carries the most error-prone part?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair is comparing answers, pointing at specific digits, debating	"I got 14702, you got 14701 — where's the difference?"
Needs intervention	One partner just copies the other's answers	Silence; "fine, whatever"

Safety / sensitivity

- No timed competition between pairs publicly. Pairs are self-pacing.

Differentiation

- **Tier up:** 5×5-digit problems (printed on a separate sheet).

- **Tier down:** 3×3-digit problems only; build up.

Clean-up

- Worksheets collected (for the teacher to spot-check before Day 5).

Activity 02 — Near-Base Mental Addition

Module: Bindu Paddhati · **Band:** Middle · **Day:** 7 · **Time:** 20 min

Type

Quick-fire mental math drill, individual + paired

Goal

Students add 2-digit numbers ending in 9, 19, 29, 49, 99 mentally using the “add the base, subtract the deficiency” pattern, in under 3 seconds per problem.

Materials per pair

- 1 printed drill sheet (20 problems below)
- 1 stopwatch

Instructions (student-facing handout)

Near-Base Mental Drill — work in pairs.

Round 1 (5 min, individual): Solve as many of the 20 problems as you can on the sheet. NO writing intermediate steps — write only the answer.

Round 2 (5 min, paired): Partner-A reads a problem aloud. Partner-B answers in under 3 seconds. Swap. Continue alternating through all 20.

Round 3 (5 min, mixed): Teacher displays 10 new problems on the board. Class shouts the answer in chorus.

Round 4 (5 min, quick-fire): Teacher calls a 2-digit number ending in 9 or 8, then any other 2-digit number; students whisper-answer to their neighbour.

Drill sheet (20 problems)

- | | |
|------------|-------------|
| 1. 33 + 9 | 11. 27 + 19 |
| 2. 47 + 9 | 12. 56 + 29 |
| 3. 64 + 19 | 13. 35 + 39 |
| 4. 28 + 19 | 14. 73 + 49 |
| 5. 55 + 48 | 15. 81 + 19 |
| 6. 36 + 38 | 16. 42 + 28 |
| 7. 19 + 19 | 17. 65 + 99 |
| 8. 72 + 28 | 18. 88 + 12 |
| 9. 46 + 39 | 19. 54 + 47 |
| 10. 91 + 9 | 20. 33 + 88 |

Answer key (teacher copy)

1. 42 2. 56 3. 83 4. 47 5. 103 6. 74 7. 38 8. 100 9. 85
10. 100
11. 46 12. 85 13. 74 14. 122 15. 100 16. 70 17. 164 18. 100 19. 101
20. 121

Discussion prompts (after Round 4)

- Which type of problem was easiest? (Most students say +9 or +19 — single digit deficiency.)
- Which type took longer mental processing? (Often +48, +99 — bigger numbers requiring base-100 thinking.)
- Anyone discover a *new* shortcut while doing this?

Safety / sensitivity

- Same as Activity 01 — no public competition.

Differentiation

- **Tier up:** 3-digit near-base additions ($234 + 99$, $567 + 199$, etc.).
- **Tier down:** Stick to +9 and +19 problems for today. Build up.

Clean-up

- Worksheets kept by students for revision.

Project Brief

Project Brief — Module 5 Dot Addition (Middle)

Work sessions: Day 8 + Day 9 · **Presentation:** Day 10 · **Total marks:** 40 (project 30 + presentation 10)

The project

Each pair (or individual, with teacher approval) chooses ONE of three formats:

Format 1: 60-second teaching video

Make a short video that teaches the dot method to a Grade 4 student. Constraints: - Total length: 60 seconds (± 10 sec) - Must show: dot placement, units digit identification, carry handoff for at least one example. - Cite the source (PDF + section) in the title slide or closing credit.

Format 2: Printed workbook

Produce a 20-problem workbook with the dot method, designed for a Grade 4 self-learner. Constraints: - 20 problems organised in 4 difficulty levels (5 problems each). - Each level

includes ONE fully worked example. - Answer key on the back page. - Title page with cited source.

Format 3: Comparative speed-test data project

Collect baseline times (standard algorithm) and post-training times (dot method) from 5–10 people (family, schoolmates). Constraints: - A table of names (or codes), problem solved, time using standard, time using dot method. - A plot or visual comparison. - A 1-paragraph interpretation of the data — including any surprising findings. - Cite the source PDF.

Why this matters

You're moving from *learner* to *teacher* (or *researcher*). Either role is harder than just doing the problems. The discipline of explaining or measuring forces you to understand more deeply.

What success looks like

A strong project will: - Demonstrate the dot method method accurately (no arithmetic errors). - Be pitched for the intended audience (a 9-year-old for video and workbook; a peer reader for data). - Show the pair's own thinking. - Be clearly communicated.

Working in pairs

- Pair work strongly encouraged.
- Individual allowed for the workbook or video format.
- Groups of 3 acceptable — each member must have a distinct role.

Citation rules

- At minimum **one** citation must appear in your project — visible.
- Format: <PDF filename>, §<section> (e.g. *Vedic Mathematics Batch1.pdf*, §5.4).
- See `sources.md` for the list of available PDFs.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to learn or make, and what did you actually end up with? 2. What's the most surprising thing you discovered while making this? 3. If you had three more days, what would you do differently?

Rubric

Project Rubric — Module 5 Dot Addition (Middle)

Project marks: 30 (out of 40 with presentation 10).

Criteria

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Method demonstrated correctly	Method shown completely (dot placement, units digit, carry); at least 3 worked examples with no errors.	Method shown completely; minor errors on 1 example.	Method shown but with conceptual gap (e.g. missing carry handoff).	Method incorrect or absent.	× 2
Use of sources	Cites at least one PDF section verbatim (e.g. “ <i>Vedic Mathematics Batch1.pdf</i> , §5.4”); cites it correctly.	Cites a PDF source.	Mentions a source vaguely.	No source cited.	× 1.5
Audience suitability	Pitched precisely to the target audience (Grade 4 for video; self-study for workbook; readable for data project).	Mostly suitable; minor tonal issues.	Suitable in places, off in others.	Inaccessible for target audience.	× 1.5
Originality	The project shows the pair’s own thinking — a unique example, a custom narrative, a real data set.	Clearly the students’ own work.	Largely repeated from class examples.	Copied verbatim from a source.	× 1
Communication / craft	Visuals, layout, narration are clear,	Communicates clearly. 1–2 craft issues.	Message comes through with effort.	Hard to follow.	× 1

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
	deliberate, polished. No errors in arithmetic shown.				

Maximum: 28. Project score = (sum × weights) × 30/28.

Presentation (10 marks)

Criterion	4	3	2	1	Weight
Within time (3 min)	2:45–3:00	2:30–3:15	2:15–3:30	Outside	× 1
Both partners contribute	Equal	Both speak unequally	One dominates	One silent	× 1
Clarity	Confident, audible	Audible, mostly clear	Audible only in front	Mostly inaudible	× 0.5

Maximum: 10.

Format-specific notes

- **Video projects:** the 60-second video itself is the artefact; the 3-min presentation explains the design choices (“we used this example because ...”).
- **Workbook projects:** the workbook is the artefact; the 3-min presentation walks through the cleverest 1–2 problems.
- **Speed-test data projects:** the table/chart is the artefact; the 3-min presentation interprets the data (“our 5 participants showed an average X% improvement”).